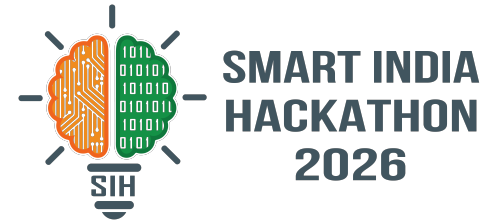


SMART INDIA HACKATHON 2026



TITLE PAGE

Problem Statement ID – PSS06

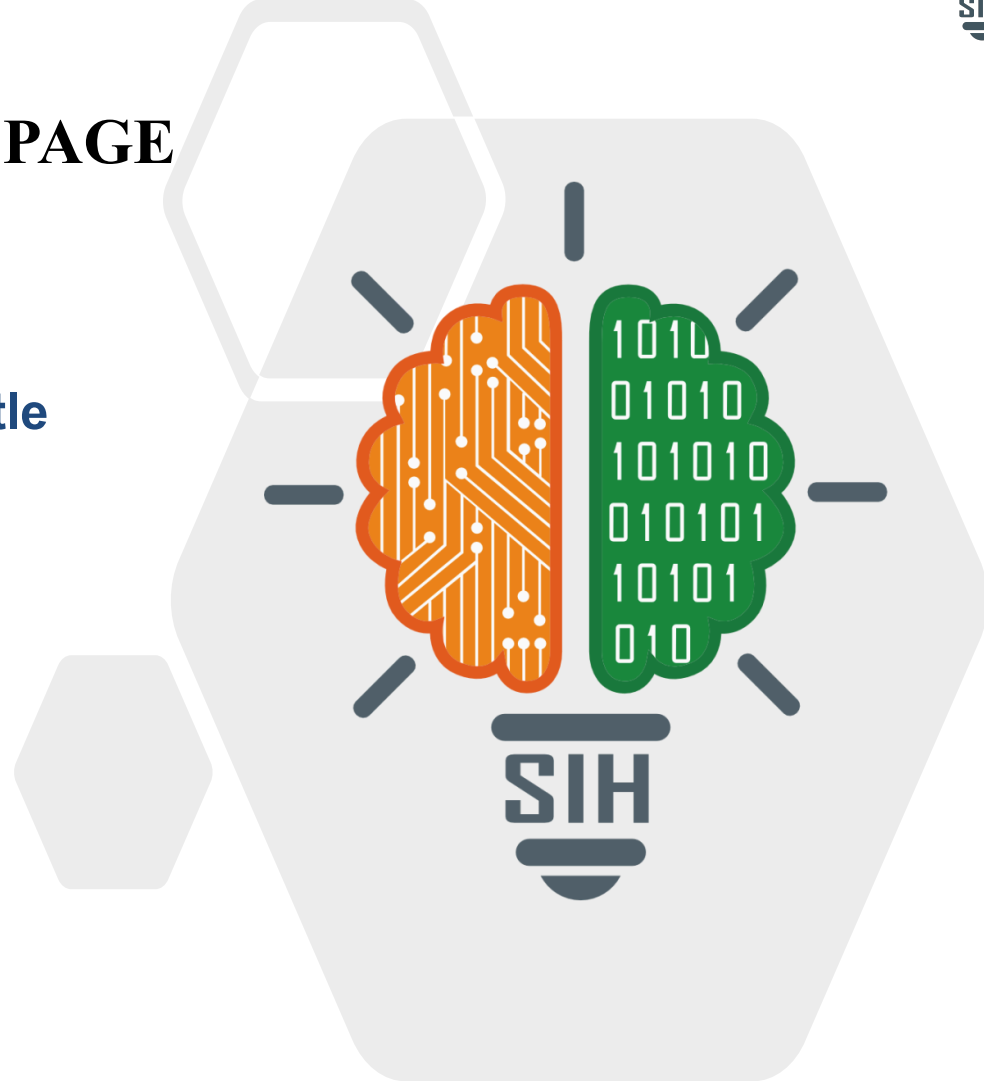
**Problem Statement Title– Automated Press Title
Admissibility Clearance (PRGI)**

Theme– Smart Automation

PS Category– Software

Team ID– 26 (SR Number)

Team Name– 404 Found



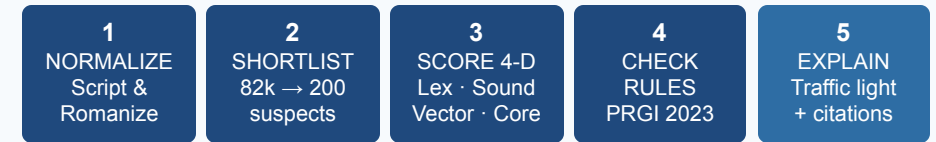
Proposed Solution (Describe your Idea/Solution/Prototype)

- **PRGI TitleGuard** — an explainable, agentic decision-support system that clears a proposed periodical title against **82,713** registered titles in **under 2 seconds**, replacing a manual cycle that today takes **25–30 days** per attempt.
- Similarity is measured on **four independent dimensions**, not plain text match — **Lexical** (pg_trgm + RapidFuzz), **Phonetic** (Double Metaphone), **Cross-lingual Semantic** (self-hosted multilingual embeddings over pgvector/HNSW) and **Core-Root** (media filler-word stripping).
- A **separate deterministic rule engine** checks statutory admissibility in plain code — never an LLM — so the same title always yields the same verdict, clause by clause.
- The **Agentic Title Studio** (LangGraph: Interviewer → Generator → Verifier → Ranker) proposes alternative titles that are **pre-verified to pass** — no unverified suggestion is ever shown.

How it addresses the problem

- › Catches the collisions manual checking misses: *Times India* vs *India Times*, *Jaagran* vs *Jagran*, *Dainik Samachar* vs *Daily News*, *The Vidarbha Daily Express* vs *Vidarbha Patrika*.
- › Ends officer-to-officer inconsistency with one standardised, reproducible score — and returns a specific, actionable reason instead of a silent rejection that costs the applicant ~₹1,000 and a full restart.

5-STAGE VERIFICATION PIPELINE



Verdict: **APPROVED** · **MANUAL REVIEW** · **REJECTED**

— always a recommendation, never a determination

Innovation and uniqueness of the solution

- › **Retrieve-then-generate citations:** the guideline clause is fetched verbatim from the actual PRGI text before any wording is generated — a fabricated citation is impossible by construction.
- › **Two purpose-built mechanisms, not one classifier:** fuzzy similarity and deterministic statute are structurally different failures and are checked separately.
- › **Suggestions that already passed:** generated alternatives re-enter the same 5-stage pipeline before they are ever displayed.

Technologies to be used

- **Backend:** Python 3.11, FastAPI (native async, Pydantic-validated contracts, auto-generated OpenAPI docs).
- **Data & search:** PostgreSQL 16 with **pg_trgm** (trigram lexical) and **pgvector + HNSW** (semantic) — relational metadata and embeddings on the same row, one system, one query.
- **AI/ML:** RapidFuzz (C++ lexical), Double Metaphone (phonetic), self-hosted multilingual Sentence-Transformers (LaBSE-class) for cross-lingual meaning, RAG over the guideline corpus for explanations.
- **Agents:** LangGraph — explicit nodes, explicit edges, inspectable state, one bounded retry loop.
- **Frontend:** Next.js + TypeScript + Tailwind CSS; Zod schemas generated from the same API contract as the backend Pydantic models.
- **Quality & delivery:** Pytest (one test per rule), Playwright end-to-end, Docker Compose, GitHub Actions.

Methodology and process for implementation

LAYER 3 — APPLICATION & AGENTS

Verification Console
Next.js / TS / Tailwind

FastAPI Gateway
/v1/verify orchestrator

Agentic Studio
Interviewer→Gen→Verify→Rank

LAYER 2 — AI/ML & STATUTORY RULES

4-D Similarity Engine
Lexical · Sound · Semantic · Core

Deterministic Rule Engine
PRP Act 2023 + citations

LAYER 1 — DATA & CANDIDATE SEARCH

PostgreSQL + pgvector HNSW
82,713 titles · trigram + vector indexes · guideline clause store

REQUEST PATH — EVERY VERIFICATION

- **1 Normalize** — detect Devanagari/Bengali/Tamil/Telugu/Gujarati/Urdu, romanize to standard phonetic tokens.
- **2 Shortlist** — parallel trigram + HNSW query narrows 82,713 titles to ~200 suspects.
- **3 Score** — each suspect scored 0–100 on all four dimensions in one batch, not one round-trip per candidate.
- **4 Rules** — character limits, generic single-word ban, commercial/matrimonial catalogue ban, URL syntax, Emblems & Names Act.
- **5 Explain** — retrieve the verbatim clause, then phrase it; emit verdict, evidence, and an exportable officer memorandum.

Every request is traced end-to-end across all five stages with per-stage timing against the <2 s target.

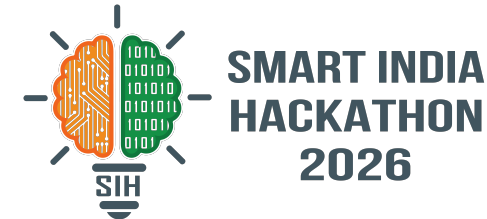
Analysis of the feasibility of the idea

- **The data already exists and is already loaded.** 82,713 real PRGI title records (`title_master.csv`) are in PostgreSQL with trigram and vector indexes built; trigram search has been validated against live queries and correctly surfaces genuine close matches.
- **The workload is small by modern standards.** 82,713 vectors is well inside what pgvector/HNSW serves sub-second on commodity hardware — no GPU, no dedicated vector database, no per-query billing, no extra network hop.
- **Self-hostable and offline-capable end to end.** Models run locally, so no title data leaves government infrastructure and there is no third-party terms-of-service dependency for a public service.
- **Reproducible from a clean machine** — a 30-second local path (`./scripts/dev.sh`) and a full Docker Compose stack, with no undocumented manual steps.
- **It fits the existing chain.** The system slots in at the first gate (Applicant → Press Sewa Portal → District Magistrate → PRGI) without replacing any statutory step.

Potential challenges and risks — and the strategy for each

Challenge / risk	Strategy for overcoming it
No ground-truth historical rejection data, so precision/recall cannot yet be proven against real outcomes.	Publish accuracy as an estimate, not a claim; keep a human officer in the loop for every ambiguous case until real evaluation data exists.
Multilingual and phonetic models are weaker on proper nouns and lower-resource languages.	Never auto-reject on a borderline similarity score alone — a rejection requires rule-engine corroboration.
Transliteration between scripts is inherently not one-to-one.	Accept as a bounded, documented limitation; run phonetic matching on the romanized form rather than chasing a “perfect” transliterator.
“Confusingly similar” is a legal judgment, not a numeric threshold — the system could be treated as more authoritative than intended.	Product and interface consistently frame every output as a recommendation; final legal determination stays with the PRGI officer.
An unconverted guideline clause is a rule the engine silently cannot check.	Track rule coverage explicitly against the source document so gaps are visible in the interface, not silent.

IMPACT AND BENEFITS



25–30 days
→ **<2 s**

Verification latency, warm

82,713

Registered titles checked every time

4-D

Independent similarity signals per match

≥95%

Shortlist recall target

100%

Verdicts traceable to a cited clause

Potential impact on the target audience

- **Applicants** get an answer in seconds instead of weeks, and a specific reason instead of silence — removing the ~₹1,000 cost and full restart that every blind rejection currently forces.
- **PRGI officers** get a standardised 4-D score, a review docket, and one-click *Endorse & Approve / Issue Rejection Order* actions, each producing a recorded decision artifact — so the outcome no longer depends on which officer picked up the file.
- **Regional and vernacular publishers** are served in their own script: multi-script input, romanization preview, and cross-lingual matching mean a Devanagari or Tamil applicant is not disadvantaged.

Benefits of the solution

- **Social** — equal, transparent access to the press registration process regardless of language, region, or familiarity with the bureaucracy; verdicts are conveyed by text label as well as colour, so the interface holds up for colourblind users.
- **Economic** — zero per-query cost and no vendor lock-in: self-hosted models and one PostgreSQL instance on commodity hardware, so the cost of the ten-thousandth check equals the cost of the first.
- **Governance** — every rule-based rejection cites a real clause and every similarity flag names the conflicting titles and their scores; an unsupported rejection is treated as a defect, not an output.
- **Environmental** — a paperless, resubmission-free first gate removes repeated physical filing cycles from the approval chain.

Scale of the benefit: the PRP Act, 2023 puts every periodical title in India through this one gate, and every applicant waits **25–30 days** for the answer. Compressing it to **under two seconds**, with each verdict citing the clause it rests on, turns press registration's slowest step into its most auditable — authority unchanged.

Statutory sources and registry data

- [1] Office of the Press Registrar General of India, “Registered Titles Database Search,” PRGI, Ministry of Information & Broadcasting, Govt. of India, 2026. [Online]. Available: <https://prgi.gov.in/registration-title-details>
- [2] Government of India, “The Press and Registration of Periodicals Act, 2023,” *The Gazette of India*, Legislative Department, Ministry of Law and Justice, Feb. 2024. [Online]. Available: https://prgi.gov.in/sites/default/files/2024-02/press_and_registration_of_periodicals_act_2023.pdf
- [3] Office of the Press Registrar General of India, “Guidelines for Admissibility of Titles for Registration of Periodicals,” PRGI, Ministry of Information & Broadcasting, Govt. of India, Jul. 2025. [Online]. Available: <https://prgi.gov.in/policy/guidelines-admissibility-titles>

How each source is used

- [1] supplies the **82,713** reference title records the similarity engine scores against; [2] establishes the statutory gate the system automates; [3] is the clause-level text from which **every** deterministic rule and every citation is derived.
- [4] and [5] are the two retrieval primitives the shortlist rests on — trigram lexical matching and multilingual sentence embeddings — queried in parallel over the same store.

Standing constraint carried through every source above: the system is a decision-support tool. It makes the determination fast, consistent, and explainable — a PRGI officer always makes it.

Technical references

- [4] PostgreSQL Global Development Group, “pg_trgm — support for similarity of text using trigram matching,” *PostgreSQL 18 Documentation*, Appendix F: Additional Supplied Modules and Extensions, 2026. [Online]. Available: <https://postgresql.org>
- [5] N. Reimers and I. Gurevych, “Sentence-BERT: Sentence Embeddings using Siamese BERT-Networks,” in *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, Association for Computational Linguistics, Nov. 2019. [Online]. Available: <https://github.com/UKPLab/sentence-transformers>
- [6] Y. Malkov and D. Yashunin, “Efficient and Robust Approximate Nearest Neighbor Search using Hierarchical Navigable Small World Graphs,” *IEEE Trans. Pattern Anal. Mach. Intell.*, 2020 — the HNSW index behind sub-second shortlisting.
- [7] L. Philips, “The Double Metaphone Search Algorithm,” *C/C++ Users Journal*, vol. 18, no. 6, Jun. 2000 — dual-code phonetic encoding, chosen over Soundex for transliterated proper nouns.
- [8] P. Lewis *et al.*, “Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks,” in *Advances in Neural Information Processing Systems (NeurIPS)*, 2020 — the retrieve-then-generate ordering that makes a fabricated legal citation structurally impossible.

Project artefacts

- **Repository** — github.com/divvyesk/prgi-title-verify (PRD, architecture, API contract, database schema, rule set, front-end reference implementation).
- **Datasets** — `title_master.csv` (82,713 records) and the structured admissibility-rules corpus with section references and retrieval-ready passages.